

RESEARCH ARTICLE

Crawling under the radar: Two novel *Paulinella* species expand knowledge about the ecology and evolution of a primary plastid-containing amoeba lineage

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Abstract

The genus *Paulinella* represents a rare, independent case of primary endosymbiosis, providing a unique system to study the early stages of organelle evolution. Here, we expand current understanding of primary plastid endosymbiosis through the discovery and characterization of two novel photosynthetic amoebae, *Paulinella marae* sp. nov. and *Paulinella murrayi* sp. nov., isolated from a brackish water habitat in North Carolina, United States. Complete chromatophore genomes and mitochondrial data revealed conserved gene content but notable structural variation, including genome rearrangements and inversion events. Phylogenetic analyses uncovered significant discordance between nuclear and organelle datasets, likely driven by substitution saturation, limited taxon sampling, and differing evolutionary signals across loci. Ecological observations over multiple years indicate that both species are in low abundance but consistently present, and when coupled with hobbyist data, support the hypothesis that photosynthetic *Paulinella* species are globally distributed yet under-sampled. These results increase known species diversity within the clade from four to six and highlight the importance of integrating field-based observations with genomic approaches. Overall, this work advances *Paulinella* as a model for studying ongoing primary endosymbiosis, lineage divergence, and the ecological strategies of low-abundance microbial eukaryotes.

Abbreviations: AA, amino acid; CDS, coding DNA sequences; EGT, endosymbiotic gene transfer; ERH, endosymbiotic ratchet hypothesis; ML, maximum likelihood; PCR, polymerase chain reaction; PBS, phosphate-buffered saline; SEM, scanning electron microscopy; UFBoot, ultrafast bootstrap; WGA, whole genome amplification.

Julia Van Etten, Sungpil Han, Debashish Bhattacharya and Hwan Su Yoon contributed equally to this work.

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KEYWORDS

biodiversity, microbial ecology, *Paulinella*, photosynthesis, primary endosymbiosis, single-cell genomics

INTRODUCTION

Primary endosymbiosis is an exceedingly rare event that led to the origin of the eukaryotic cell and the subsequent evolution of photosynthesis in eukaryotes (Archibald, 2015; Löffelhardt, 2014; Reyes-Prieto et al., 2007; Sagan, 1967; Stephens et al., 2021). These processes have had immeasurable impacts on the genomic, cellular, and biological diversity on our planet. The photosynthetic *Paulinella* clade comprises rhizarian amoebae that acquired photosynthesis via primary endosymbiosis involving a free-living α -cyanobacterial cell, 90–140 million years ago (mya; Calatrava et al., 2022; Gabr et al., 2020). This is one of only two known cases of primary plastid origin, with the second putatively occurring once in the ancestor of Archaeplastida over 1.5 billion years ago and involving the capture of a β -cyanobacterium (Sibbald & Archibald, 2020). Genome size of the photosynthetic organelle in *Paulinella*, termed the “chromatophore,” is approximately six times larger than in other algal and plant plastids (Lhee et al., 2019) and appears to be undergoing further integration into host cell biology (Gabr et al., 2022). This “incomplete” state when compared to Archaeplastida offers the unique opportunity to study the fundamental processes of primary endosymbiosis and organogenesis in action, rather than retroactively, as is the case with highly derived Archaeplastida model species such as *Chlamydomonas reinhardtii*, *Arabidopsis thaliana*, and *Porphyridium purpureum*.

Investigating *Paulinella* cell biology, genomics, and natural history is, however, hindered by several key factors. To date, only four photosynthetic *Paulinella* species have been described: *P. chromatophora* (Lauterborn, 1895), *P. micropora* (Lhee et al., 2017), *P. longichromatophora* (Kim & Park, 2016), and *P. acadia* (Pardasani et al., 2025). From these species, only two nuclear genomes (*P. chromatophora* CCAC0185 and *P. micropora* KR01), which differ greatly in size and assembly quality, have been determined (Lhee et al., 2021; Nowack et al., 2016), and chromatophore genomes are available from three species (*P. chromatophora* CCAC0185; *P. micropora* KR01, FK01, and NZ27; and *P. longichromatophora*; Lhee et al., 2019). Furthermore, *Paulinella* spp. are difficult to maintain in laboratory culture and have extremely slow doubling times (5–7 days), aspects that may not reflect growth rates in their natural habitat (Gabr et al., 2022; Zhang et al., 2017). This latter trait makes it highly challenging to generate enough cells for standard approaches such as proteomics (e.g., with isolated cell fractions)

and reverse genetics. These cells are also very small (typically 10–30 μ m in diameter), adherent, motile, and in very low abundance in nature (see below), presenting issues for isolation and successful cultivation. Thus, many crucial questions remain of *Paulinella* biology and biodiversity, despite its value as a photosymbiosis model system.

In this study, we begin closing this knowledge gap by describing two new photosynthetic *Paulinella* species that were isolated from a brackish water habitat in North Carolina, United States. We report complete chromatophore genomes, mitochondrial genomes, nuclear taxonomic markers, and ecological insights from observing these cells in natural samples for 5 years. Finding two distinct species with overlapping distribution, coupled with recent anecdotal observations across dozens of countries (<https://www.inaturalist.org/observations>; Figure S1) and widespread putative detection via non-targeted barcoding efforts across the global ocean (<https://www.gbif.org/occurrence/search>; Figure S2), it is likely that photosynthetic *Paulinella* spp. have a cosmopolitan distribution. Their low abundance and lack of name recognition have led to undersampling and a dearth of understanding about *Paulinella* natural history and habitats from which to source cells for observation, experimentation, and comparative genomics work (Van Etten et al., 2023). By increasing knowledge about *Paulinella* and its distribution and shifting the focus from in vitro to in situ studies, it becomes possible to better understand the evolution of primary endosymbiosis and lineage divergence and how rare taxa persist in microbial communities (Bhattacharya et al., 2023).

MATERIALS AND METHODS

Sampling and isolation

Photosynthetic *Paulinella* cells were isolated from a brackish water sample collected on May 12, 2025, from Roanoke Sound in Nag's Head, North Carolina, United States (Figure S3; and for more sampling details, see Text S1). This is a location our team has been visiting since 2021, and after extensive observation of natural samples, we noticed two distinct cell morphologies that were recorded for future work (Figure S4). Smaller cells were observed more often either free-floating in the water (some likely shaken loose from plants when the sample was collected and transported), and the larger cells were observed more often on crust from the surface of grasses collected in the water sample from the site at the time of

sampling. Using pulled Pasteur pipettes, regular micro-pipettes, and tools made from toothbrush bristles superglued onto colored pencils, individual cells were isolated and washed in 0.2- μ m filtered ambient water and then either placed in ESM-C media with 0.22- μ m-filtered ambient water in a 1:5 ratio for culturing, phosphate-buffered saline (PBS) for single-cell genomics, or 50% ethanol for scanning electron microscopy (SEM) imaging. Effort was made to isolate cells representing both morphologies for downstream applications.

Whole genome amplification and polymerase chain reaction

Ten cells were isolated, imaged, and washed in PBS for whole genome amplification (WGA; records of measured cells and associated metadata are in [Tables S1](#) and [S2](#)). Based on light microscopy, Cells 2, 4, and 9 appeared to be of the “larger morphology”; Cells 5, 6, 7, and 10 were of the “smaller morphology”, and Cells 1, 3, and 8 could not be imaged well. Some cells (3, 7, and 10) were lost during the washing process. The PBS wash was sometimes destructive to cells (1, 2, and 8), rendering them fragile, and they broke during the attempted transfer from PBS to the lysis buffer. Cell 2 was ruptured, but a single chromatophore was reisolated and transferred to lysis buffer for the WGA. One WGA reaction was carried out for each remaining cell sample (2, 4, 5, 6, and 9) following the Qiagen REPLI-g Mini Kit protocol. DNA was stored at 4°C for 2 days, and DNA concentration and integrity were assessed using the Agilent 2200 TapeStation system. Aliquots of each sample were taken for polymerase chain reaction (PCR) and barcoding of the 16S rDNA (chromatophore) and 18S rDNA (nuclear) gene sequences; details are in [Text S2](#). Barcoded amplicons were pooled into a single library, and sequencing was completed on a MinION device using a Flongle Flow Cell (FLO-FLG114) according to the manufacturers' instructions, to assess the success of the WGA by checking for the presence of *Paulinella* DNA. Samples from Cells 2, 5, 6, and 9 were positive for a *Paulinella*-like 16S rDNA gene, and Cell 9 was positive for an 18S rDNA gene. Those four samples were retained for sequencing (see below and [Text S2](#) for sequences).

Light microscopy and scanning electron microscopy

Individual cells were isolated in the field using Amscope T340B upright and Leica DMI1 inverted microscopes and further in the lab using a Leica DMI8 inverted fluorescence microscope. Images were captured using a color camera (DMC4500) and fluorescence excitation at 640/30 and color capture of resultant fluorescence (no

specific emission filter). For SEM, previous attempts to recover cells from 4% paraformaldehyde prepared in the field or from methanol fixation followed by ethanol dehydration with hexamethyldisilazane failed to recover intact cells. Instead, two cells of different morphology were isolated, washed in 0.22- μ m filtered ambient water, followed by a wash in 50% ethanol, and then mouth-pipetted directly onto the SEM stub, which was placed in a desiccator prior to imaging. It was then coated with gold in a Cressington 108 sputter coater and observed with a Phenom ProX Desktop scanning electron microscope (Nanoscience) located at Bigelow Laboratory for Ocean Sciences, Boothbay, Maine, United States.

Sequencing assembly, validation, and annotation of organelle genomes

Four WGA extracts (from Cells 2, 5, 6, and 9) were sequenced using the PacBio HiFi platform (Biosciences of California, Inc). Chromatophore genomes of Cells 5 and 9 were assembled from PacBio HiFi reads using TIPPO v 1.3.0 (Xian et al., [2025](#)). To assemble the chromatophore genome, the relative order and orientation of the resulting contigs were aligned to the genome of the closely related species *Paulinella longichromatophora* using LASTZ v1.04.15 (Harris, [2007](#)). Based on this first draft pseudo-assembly, contigs were ordered and concatenated into an initial linear scaffold by inserting 5000bp gaps (Ns) between adjacent contigs. Raw reads were then mapped to the scaffold using minimap2 (Li, [2018](#)). Contigs were iteratively extended and merged within the scaffold using read-mapping evidence: (i) When reads consistently extended beyond contig ends (i.e., overhanging/soft-clipped segments supported by multiple reads), such extensions were considered valid, and contigs were extended accordingly, and (ii) when split or spanning reads bridged adjacent contigs, contigs were merged using the read-supported junction. This map-extend/merge procedure was repeated until no further improvement was observed. When the assembly stalled, alternative contig orders were tested by rearranging contigs followed by remapping and re-evaluation. Finally, circularization of each chromatophore genome was confirmed by raw read mapping across the circular pseudo-assembly junction.

For validation, raw reads were mapped back to each finalized chromatophore assembly using minimap2 (Li, [2018](#)). Alignments were filtered to retain high-confidence mappings ($\geq 98\%$ sequence identity and $\geq 98\%$ query coverage). Mapping depth and coverage uniformity were quantified with mosdepth (Pedersen & Quinlan, [2018](#)) to assess assembly quality. Genome annotation was done using Geneious v2025.2.2 with a custom reference database built from chromatophore genomes of photosynthetic *Paulinella* species (Lhee et al., [2019](#)). The tRNA genes were curated

with RNasease (Lang et al., 2007). Remaining open reading frames were annotated by BLASTx (Camacho et al., 2009) searches against the National Center for Biotechnology Information (NCBI) databases using default parameters, with an E-value threshold of 1e-5 and a maximum of 100 target sequences. These annotations were assigned when the top BLASTX hits showed consistent protein descriptions. Genome maps were visualized with CGView (Stothard et al., 2019). Mitochondrial genomes were assembled, validated, and annotated using the same workflow described above.

Gene content comparison and rearrangement analysis

Chromatophore genomes from Cells 5 and 9 (one of each morphology) were analyzed together with chromatophore genomes from four taxa of photosynthetic *Paulinella*. To compare protein-coding gene content, protein-coding sequences were extracted from each chromatophore genome and clustered into orthogroups using OrthoFinder (Emms & Kelly, 2019). Orthogroup membership and gene-count profiles were then used to compare gene content among phototrophic *Paulinella* chromatophore genomes. To analyze structural rearrangements, collinearity among chromatophore genomes was examined by performing pairwise alignments with LASTZ v1.04.15 (Harris, 2007) and visualizing conserved blocks in AliTV (Ankenbrand et al., 2017). The same collinearity analysis workflow was applied to the mitochondrial genomes.

Phylogenetic analysis

To resolve the phylogenetic relationship between the two novel brackish-water *Paulinella* isolates, phylogenies were reconstructed using the 18S rDNA genes as well as chromatophore and mitochondrial coding DNA sequences (CDSs). For the 18S rDNA gene phylogeny, PacBio HiFi raw reads generated for Cells 5 and 9 were searched by BLASTn using the 18S rDNA gene sequence of *P. micropora* KR01 as the query (Camacho et al., 2009). These BLASTn searches were performed with an E-value cutoff of 1e-10 and otherwise default parameters, and matching reads were recovered as 18S rDNA gene sequences. Redundant sequences were subsequently removed using CD-HIT (Fu et al., 2012). The resulting sequences were combined with an Imbricatea 18S rDNA gene reference dataset, aligned using SSU-Align, and masked with SSU-Mask to exclude positions that were considered likely to be noisy or ambiguously aligned (Nawrocki, 2009). A maximum-likelihood (ML) phylogeny was inferred using IQ-TREE 2 under the TN+I+R2 model,

which was selected as the best-fit substitution model by ModelFinder (Minh et al., 2020). Branch support was assessed using 1000 ultrafast bootstrap (UFBoot) replicates. The 28S rDNA gene phylogeny was processed in the same manner. However, sequence alignment was done using MAFFT (Katoh, 2002) and trimmed using BMGE (Criscuolo & Gribaldo, 2010). The TIM2+F+R2 model was selected as the best-fit substitution model by ModelFinder. For the chromatophore phylogeny, a dataset including the assembled chromatophore genomes of Cells 5 and 9 was analyzed. Orthologous genes were inferred and clustered using OrthoFinder (Emms & Kelly, 2019). In total, 42 genomes comprising seven photosynthetic *Paulinella* taxa and representative cyanobacterial genomes were included, and *Gloeobacter violaceus* PCC 7421 was used as the outgroup. The corresponding protein sequences were aligned with MAFFT (Katoh & Toh, 2008). A concatenated alignment was constructed from 406 orthologous genes and analyzed using IQ-TREE 2 under a partitioned ML framework (Minh et al., 2020). The best-fit substitution model for each gene partition was selected by ModelFinder, and branch support was evaluated using 1000 UFBoot replicates. For the mitochondrial phylogeny, the dataset comprised six photosynthetic *Paulinella* taxa, including Cells 5 and 9, three heterotrophic *Paulinella* taxa, one Stramenopile taxon, and eight Rhizaria taxa. Nineteen orthologous genes were translated to amino acid sequences, aligned with MAFFT (Katoh & Toh, 2008), and trimmed using BMGE (Criscuolo & Gribaldo, 2010). Using the same workflow as described above, the trimmed alignments were concatenated and analyzed in IQ-TREE 2 (Minh et al., 2020).

Phylogenetic discordance was quantified using PhyParts (Smith et al., 2015). To evaluate incongruence among gene trees based on chromatophore-derived phylogenetic signal, the concatenated chromatophore gene tree was used as the reference tree. A total of 378 orthologous genes were analyzed. For each gene, an independent gene tree was inferred, and concordant versus conflicting bipartitions (splits) were assessed at each node of the reference tree. All gene trees were rooted using the outgroup *Gloeobacter violaceus* PCC 7421 prior to downstream analyses. PhyParts outputs were visualized as pie charts. In addition, gene-specific topologies were classified by applying a custom Python script to the 378 gene trees. For each gene tree, bootstrap values for internal branches within the *Paulinella* clade were extracted, and gene trees were filtered by requiring bootstrap support of ≥ 50 for those branches; the frequencies of the resulting gene-tree topologies were then calculated. Using this approach, the four most prevalent topologies among photosynthetic *Paulinella* lineages repeatedly recovered across chromatophore genes were identified. The same workflow was applied to the 19 mitochondrial gene trees.

Assessment of substitution saturation in the *rbcL* gene

To investigate substitution saturation in the *rbcL* gene, phylogenetic analyses were done using codon-position-specific nucleotide datasets and the translated amino acid (AA) sequences. A total of 24 taxa were included, comprising seven photosynthetic *Paulinella* and 17 representative cyanobacteria. The *rbcL* sequences were aligned in a codon-aware manner with MACSE v2.07 (Ranwez et al., 2018). From the full nucleotide alignment, three datasets were generated using a custom Python script: all codons (NT123), the first and second codon positions (NT12), and only the second codon position (NT2).

Maximum-likelihood phylogenetic analysis was done in IQ-TREE 2 using the NT123, NT12, NT2, and AA datasets, with branch support assessed with 1000 ultrafast bootstrap replicates (Minh et al., 2020). Substitution saturation was evaluated using saturation plots (Philippe & Forterre, 1999). For this analysis, the NT123 and AA datasets and each individual codon position (NT1, NT2, NT3) from the seven photosynthetic *Paulinella* taxa were used to infer an ML tree with IQ-TREE 2 (Minh et al., 2020). Using the R package ape, observed pairwise distances were calculated from the alignments as uncorrected *p*-distances, whereas ML-inferred pairwise tree distances were obtained as patristic distances from the branch lengths of the corresponding ML trees (Paradis & Schliep, 2019). Saturation plots and linear regressions were generated using a custom R script.

RESULTS

Species descriptions

Taxonomy, via AlgaeBase (Guiry & Guiry, 2024)

Empire: Eukaryota, Chatton, 1925

Kingdom: Chromista, Cavalier-Smith, 1981

Subkingdom: Harosa (Supergroup SAR), Cavalier-Smith, 2010

Infrakingdom: Rhizaria, Cavalier-Smith, 2002

Phylum: Cercozoa, Cavalier-Smith, 1998

Superclass: Ventrifilosa, Cavalier-Smith, 2012

Class: Imbricatea, Cavalier-Smith, 2003

Order: Euglyphida, H.F. Copeland, 1956

Family: Paulinellidae, De Saedeleer, 1934

Genus: *Paulinella*, Lauterborn, 1895

Species: *Paulinella marae*, Van Etten, 2026

Species: *Paulinella murrayi*, Van Etten, 2026

Paulinella marae sp. nov. J. Van Etten, D. Bhattacharya, J. A. Burns, S. Han, H. S. Yoon

Diagnosis (*Paulinella marae*)

Tests are elongate, ~15–19.1 µm long (avg: 17.32 µm) and 12–15.9 µm wide (avg: 13.44 µm); test with five columns of scales, each column with nine to 10 rows of non-overlapping silicious, curved rectangular scales with slightly rounded corners. Scales overlap in a left-handed manner, counterclockwise. Aperture opening at narrower end of cell (anterior) surrounded by five collar scales of unequal sizes. Two primary collar scales flank the operculum, the width of which is approximately 3.5 µm. Scales at anterior and posterior poles of the test are smaller in length than in the middle region of the test. The posterior view has a diameter of approximately 12–13 µm where the five-fold symmetry due to the five scale rows is most pronounced and flower-like. There are punctae on the external scale surfaces at the posterior end of the tests. Cells have blue-green, “sausage-shaped,” photosynthetic chromatophores, each approximately 2.8 µm in diameter (circumference of ends is ~8.7 µm). The straight path between ends of a single chromatophore is ~13.7 µm, and the curved length is ~14.8 µm. Following cell division (where the parent cell begins to place scales outside its test to build the new test), cells have one chromatophore, and prior to division, they have two. During movement, filopodia are extremely thin and difficult to see unless viewed in differential interference contrast and can extend upwards of 60 µm or four times the length of the cell. Molecular features: The chromatophore genome is 970,001 bp in length and comprises 861 CDSs with six rRNAs and 42 tRNAs; GC content is 36.5%. Gene sequences: Sequences of nuclear 18S and 28S rDNA genes were extracted from raw sequencing reads in SRR38183187; the 18S rDNA gene accession is PZ578613 and the 28S rDNA gene accession is PZ582844. Chromatophore and mitochondrial genomes of *Paulinella marae* are in GenBank under accession numbers PZ319163 and PZ318597, respectively, and the *rbcL* accession is PZ584713.

Etymology. This name honors Mara Van Etten, who is a co-discoverer. The specific epithet (*mara*) is a Latin-derived name, meaning “sea.”

Holotype. Scanning electron microscopy stub deposited at the Chrysler Herbarium of Rutgers University (New Jersey, United States), barcode no. CHRB 0145024. Collection of one specimen (along with one *Paulinella murrayi* sp. nov. cell). Sample was collected from a roadside dock on Roanoke Sound, in Nag's Head, North Carolina, United States (35°54'10" N, 75°36'15" W). Sample was first collected on June 3, 2021, by J. Van Etten. Cells used for molecular

work and SEM visualization were collected on May 12, 2025. Type habitat: Brackish wetland (salinity 10–14), some free-floating cells but most associated with black needlerush grass or watergrasses (and associated diatom-rich biofilms) growing in the sand a few feet offshore. Distribution: Unknown.

Paulinella murrayi sp. nov. J. Van Etten, D. Bhattacharya, J. A. Burns, S. Han, H. S. Yoon

Diagnosis (*Paulinella murrayi*)

Tests are elongate, 13.1–15.3 μm long (avg: 14.41 μm) and ~11–12.8 μm wide (avg: 11.94 μm); test with five columns of scales, each column with eight to nine rows of non-overlapping silicious, curved rectangular scales with slightly rounded corners. Scales overlap in a right-handed manner, clockwise. Aperture opening at narrower end of cell (anterior) surrounded by three to five collar scales of unequal sizes. Two primary collar scales flanking the operculum described in other species are missing. Scales at anterior and posterior poles of the test are smaller in length than in the middle region of the test. There are punctae on the external scale surfaces at the posterior end of the tests and scales with dark pore-like circles throughout, surrounded by denser areas that appear lighter by SEM. The posterior view has a diameter of approximately 12 μm where the five-fold symmetry, due to the five scale rows, is most pronounced and flower-like. Cells have blue-green, “sausage-shaped,” photosynthetic chromatophores, each approximately 2.4 μm in diameter (circumference of ends is ~7.4 μm). The straight path between ends of a single chromatophore is ~13.7 μm , and the curved length is ~14.8 μm . Following cell division (where the parent cell begins to place scales outside its test to build the new test), cells have one chromatophore and prior to division, they have two. During movement, filopodia are extremely thin and difficult to see unless viewed in differential interference contrast and can extend upwards of 60 μm or four times the length of the cell. Molecular features: The chromatophore genome is 980,939 bp in length and comprises 866 CDSs with six rRNAs and 42 tRNAs; GC content is 37.5%. Gene sequences: Sequences of nuclear 18S and 28S rDNA genes were extracted from raw sequencing reads in SRR38183186; the 18S rDNA gene accession is PZ578619, and the 28S rDNA gene accession is PZ582845. Chromatophore and mitochondrial genomes of *Paulinella murrayi* are in GenBank under accession numbers PZ319162 and PZ318594-6, respectively, and the *rbcL* accession is PZ584714.

Etymology. The specific epithet (*murrayi*) is a Gaelic-derived surname, meaning “settlement by the sea,”

“seashore,” or “mariner,” as well as “strength” and “sea warrior.”

Holotype. Scanning electron microscopy stub deposited at the Chrysler Herbarium of Rutgers University (New Jersey, United States), barcode no. CHRB 0145025. Collection of one specimen (along with one *Paulinella marae* sp. nov. cell). Sample was collected from a roadside dock on Roanoke Sound, in Nag's Head, North Carolina, United States (35°54'10" N, 75°36'15" W). Sample first collected on June 3, 2021, by J. Van Etten. Cells used for molecular work and SEM visualization were collected on May 12, 2025. Type habitat: Brackish wetland (salinity 10–14), mostly free-floating cells but some associated with watergrasses (and associated diatom-rich biofilms) growing in the sand a few feet offshore. Distribution: Unknown.

Occurrence of *Paulinella murrayi* sp. nov. and *P. marae* sp. nov.

Cells 5 and 9 of *Paulinella* spp., referred to hereafter as *Paulinella murrayi* and *Paulinella marae*, respectively, were isolated from a roadside brackish water marsh in Nag's Head, North Carolina, United States (Figure S3). The water salinity ranged from 10 to 14, and pH was ~5.98. These conditions likely fluctuate year round due to the in/outflow of nearby ocean water (Jia & Li, 2012). This site was visited one to two times each year starting in 2021 in May, June, July, or August, and *Paulinella* cells were rare but observed each time, with higher abundances noted in early June. Cells of both morphologies were often observed in the same pipette, suggesting an overlapping distribution of both species, although *Paulinella* cells are so rare that there was never more than one cell observed in a given drop of water during examination (and many drops had none). Years of isolating cells from this site allowed us to retrieve the ~22 cells used for this study (10 for WGA, six for SEM, and six for culture), which were isolated over the course of 1 week in May 2025. Most cells were lost in the process; four, however, were successfully processed for WGA and two for SEM. Attempts to establish a long-term axenic culture have not yet been successful, although the cells survived and divided in a combination of filtered ambient water and media, sometimes for up to 1 month.

Morphology and behavior show two distinct populations

Between 2021 and 2024, we observed two distinct morphotypes in natural samples. It is difficult to quantify cell-specific differences using light microscopy in the field; however, some cells (*Paulinella marae*-type

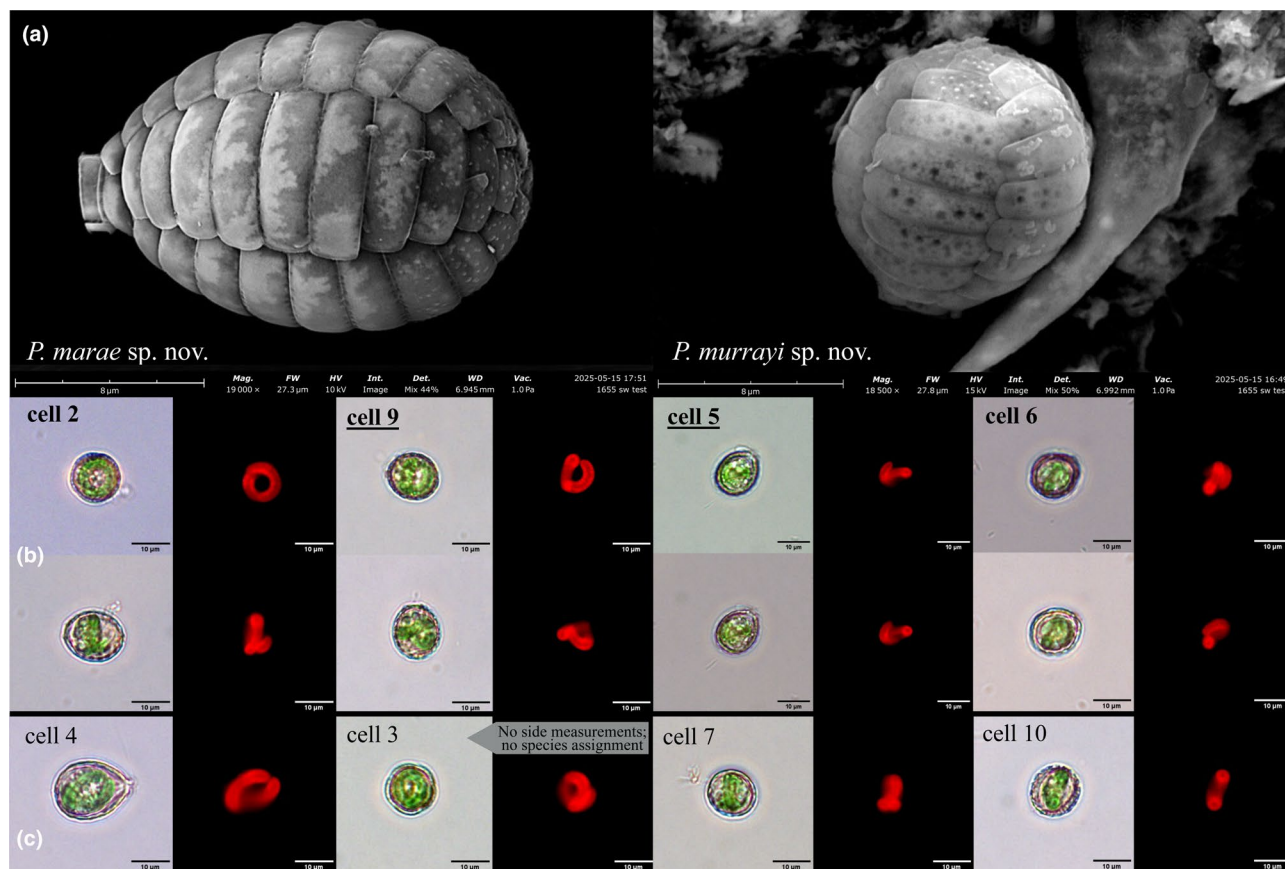


FIGURE 1 Images of *Paulinella marae* sp. nov. (left) and *P. murrayi* sp. nov. (right). The top panel (a) shows SEM images. The middle panel (two rows of images, b) shows the four cells that were successfully sequenced under the SEM image; we hypothesize that they correspond to the same species, that is, Cells 2 and 9 and the SEM image above them were used to describe *P. marae*, and Cells 5 and 6 and the SEM image above them were used to describe *P. murrayi*. Each image has two views, brightfield (left) and chlorophyll autofluorescence of chromatophores under excitation of 640/30 nm (right). These cells are each shown at two angles to better visualize the chromatophores. We tried to photograph each cell as flat as possible to standardize how we generated length and width measurements, but since these were live cells, not every angle is identical across all cells. For example, Cells 2 and 9 were clearly visually the larger morphology (similar to Cell 4), but we could not photograph them at the proper angle—this also applies to the *P. murrayi* SEM image—so the measurements are underestimated and are thus excluded from calculations of the average measurements for each species (Table S1). The bottom panel (c) shows four cells that were isolated and imaged for WGA but did not yield DNA. More information for each cell is in Table S1.

morphology) appeared larger, smoother, and more transparent with more scales per row and well-defined operculum scales. In contrast, other cells (*P. murrayi*-type morphology) appeared smaller and with a rougher texture and fewer scales per row (perhaps lacking the operculum scales), creating a more opaque appearance (Figure S4). These distinguishing features were confirmed with SEM imaging, which demonstrated that the cells differed considerably in morphology (Figure 1, Figures S5, S6). Furthermore, we noted different handedness, that is, the direction in which scales overlapped was clockwise in *P. murrayi* and counterclockwise in *P. marae* (Figure S7). This feature is inherited (Nomura et al., 2014, 2023), and the differences appeared consistent within but not between species (see Table 1 and SEM images in Yoon et al. (2009), Reyes-Prieto et al. (2010), Kim & Park (2016), Lhee et al. (2017), Pardasani et al. (2025)), making this a putative new diagnostic character to consider. Morphological differences

are quantified in more detail in the species descriptions above and in Table 1. Behaviorally, the larger cells were sometimes observed adhering to the substrate, using their filopodia to create a stalk-like, “suctioning” structure (Figure S8). The smaller cells were typically observed free-floating. Both species, when moving, can extend their filopodia to over four times the length of the test (cell body; Figures S9, S10).

Molecular data support recognition of two distinct *Paulinella* species

The striking physical differences between the two novel *Paulinella* populations led us to investigate whether differences were robust enough to consider the two morphotypes as distinct species. In protists, 1% sequence divergence in the 18S rDNA gene is often used to arbitrarily (i.e., without crossing data)

TABLE 1 A comparison of morphological characters for photosynthetic *Paulinella* species and strains with data from the tables in Lhee et al. (2017) and Pardasani et al. (2025) and including *P. marae* sp. nov. and *P. murrayi* sp. nov.

	<i>P. marae</i> sp. nov.	<i>P. murrayi</i> sp. nov.	<i>P. acadia</i>	<i>P. longichromatophora</i>	<i>P. micropora</i> KR01	<i>P. micropora</i> FK01	<i>P. micropora</i> NZ-27	<i>P. chromatophora</i> CCAC0185	<i>P. Chromatophora</i> NZ-13.11
Habitat	Brackish water (10–14 ppt)	Brackish water (10–14 ppt)	Marine sediment (20–27 psu)	Marine sediment (29–31) psu	Fresh water	Fresh water	Fresh water	Fresh water	Fresh water
Chromatophore morphology	Sausage-shaped	Sausage-shaped	U-shaped	U-shaped, variable	Sausage-shaped	Sausage-shaped	Sausage-shaped	Sausage-shaped	Sausage-shaped
Length (µm)	15.9–19.1	13.1–15.3	28.8–31.9	27–35	11.5–13.8	15–17	15.8	23–27	25.8–29.4
Width (µm)	11.8–15.9	11.1–12.8	21.5–23.9	14–19	9.1–10.7	10–11	12.7	16–20	19.4–19.8
Length/Width	1.301	1.207	1.336	1.878	1.277	1.523	1.244	1.388	1.408
Oral scales	5	5	5	5	5	5	5	3	3
Number of columns	5	5	5	5	5	5	5	5	5
Scales per column	9–10	8–9	11	10–12	8–9	10–11	10	12–14	12–14
Scale features	Smooth ext.; basal pustules	Pores ext.; basal pustules	Fine pores ext.	Smooth ext.; sieve-plate pores int.	Fine pores ext.; basal pustules; sieve-plate pores int.	Fine pores ext.; basal pustules; sieve-plate pores int.	Fine pores ext.; basal pustules; sieve-plate pores int.	Pustules ext.; sieve-plate pores int.	Fine pores ext.; pustules on bottom of ext.
Handedness	Left-handed, counterclockwise	Right-handed, clockwise	Right-handed, clockwise	Right-handed, clockwise	Right-handed, clockwise	Right-handed, clockwise	Left-handed, counterclockwise	Left-handed, counterclockwise	Left-handed, counterclockwise

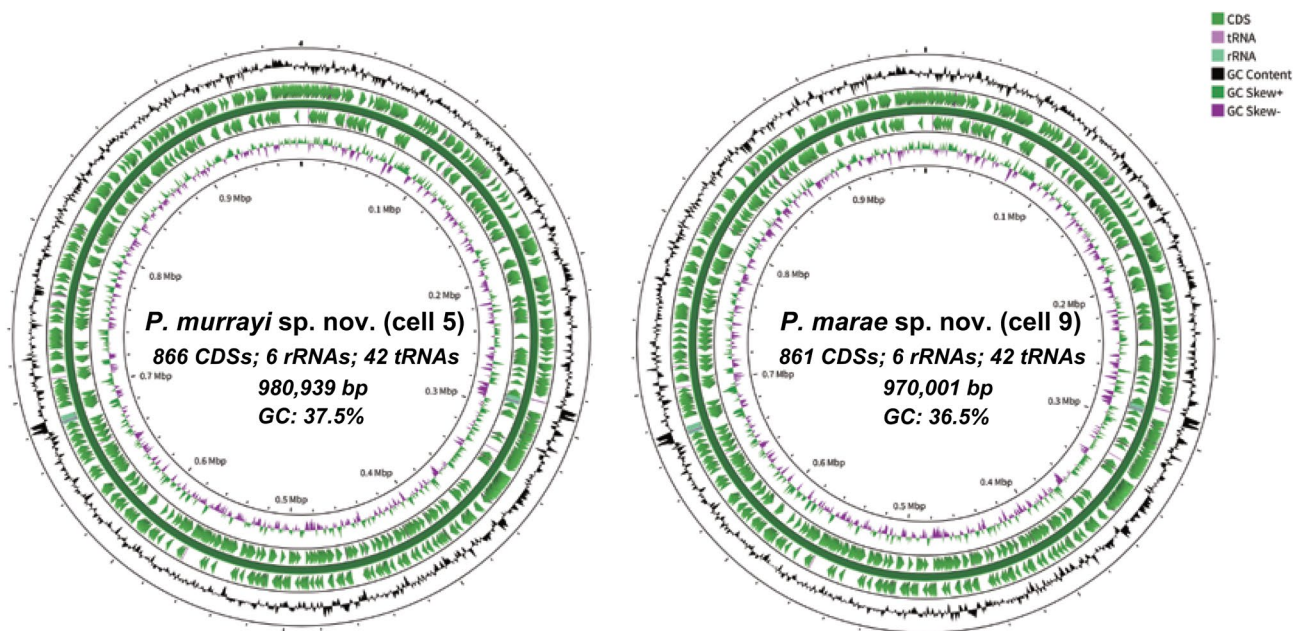


FIGURE 2 Chromatophore genome maps of novel brackish water *Paulinella* species. *Paulinella murrayi* sp. nov. (Cell 5) is on the left and *P. marae* sp. nov. (Cell 9) is on the right. Maps were visualized using CGView.

differentiate species including within the *Paulinella* clade (Jerome et al., 1996; Katz et al., 2005; Lhee et al., 2017), although thresholds for species delimitation in microbes that may lack an obvious sexual cycle are nebulous and handled differently in disparate lineages (Adl et al., 2007; Bachy et al., 2013; Caron et al., 2009; Jamy et al., 2026). The most recent species description within the *Paulinella* clade is *P. acadia*, which has a pairwise percent similarity of 98.3%–99.2% for the 18S rDNA gene sequence and 96.7%–97.2% for the 28S rDNA gene sequence when compared to its sister taxon *P. longichromatophora* (Pardasani et al., 2025). These two organisms have different morphologies and were isolated from opposite sides of the Pacific Ocean (intertidal sandy sediments from Gomso Bay, Korea, for *P. longichromatophora* and a brackish water beach near Vancouver, Canada, for *P. acadia*). It is clear they are different species despite sharing ~99% similarity of the 18S rDNA gene sequence. In our analysis, Cells 5 and 6 (*P. murrayi*) and Cell 9 (*P. marae*) had 18S rDNA gene sequences that shared 98.47%–98.66% identity and Cells 5 and 9 shared 96.87% identity in their 28S rDNA gene sequence (Table S3). These cells co-occur but have different morphological features, a significant structural inversion in the chromatophore genome (Figures 2, 3), highly diverged mitochondrial gene order (Figure 3), and divergent *rbcL* gene sequences (89.24% similarity). Despite sympatry, the North Carolina species are more distinct from one another with respect to rDNA gene identity than are *P. acadia* and *P. longichromatophora*.

Organelle genomes show structural variation between species

Four cells (2, 5, 6, and 9), obtained by single-cell isolation, were whole-genome amplified and sequenced using the PacBio HiFi sequencing platform. To minimize mis-assemblies caused by chimeric reads (Lu et al., 2023), contigs assembled with a HiFi-optimized assembler were further manually extended and merged iteratively using read-mapping evidence. This step allowed us to obtain complete circular chromatophore genomes for each cell/species (Figure 2). The same strategy was applied to the mitochondrial genome assemblies. *Paulinella murrayi* (Cell 5) yielded three partial mitochondrial contigs, whereas *P. marae* (Cell 9) yielded a single complete circular mitochondrial genome. The organellar sequencing data from Cells 2 and 6 were not complete enough to assemble into circular genomes, so these cells were excluded from downstream organelle genome comparisons and phylogenetic analyses. The resulting organelle assemblies for Cells 5 and 9 were supported by high read-mapping coverage (chromatophore mean coverage: 2975× for Cell 5 and 1014.89× for Cell 9; mitochondria mean coverage: 172.77–497.08× for Cell 5 and 692.72× for Cell 9; Table S4).

The organelle genomes obtained here did not differ substantially from those of other photosynthetic *Paulinella* in overall genome features (Table S5). For the chromatophore, the genome sizes of *P. murrayi* and *P. marae* were 980,939 bp and 970,001 bp, respectively, comparable to previously reported photosynthetic *Paulinella* species: 976,991 bp in *P. micropora* KR01,

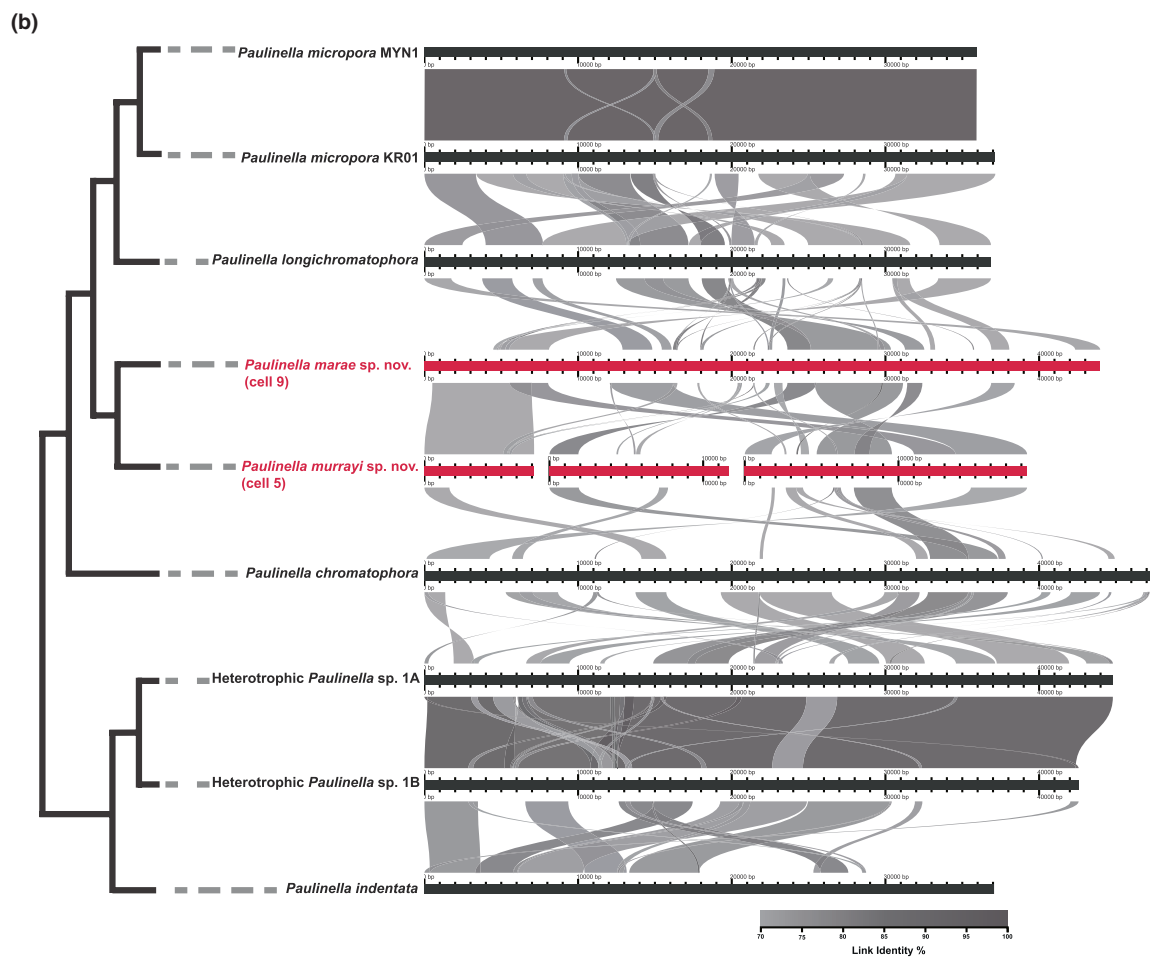
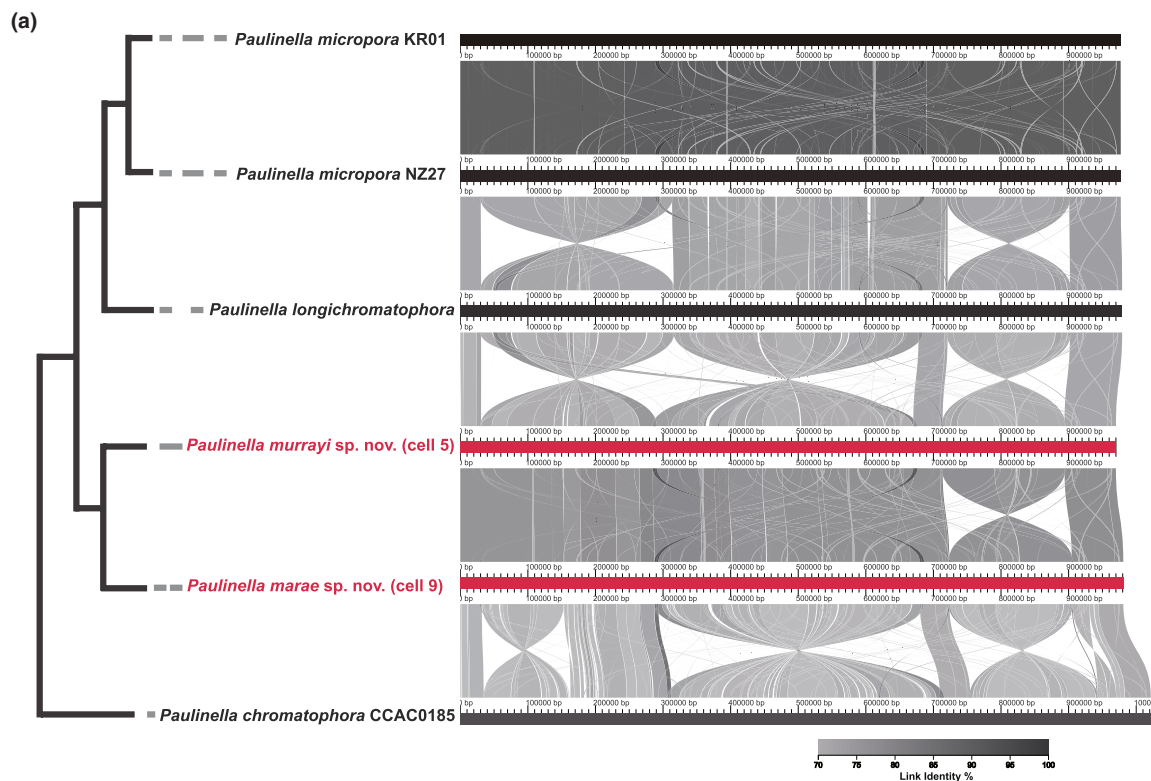


FIGURE 3 Collinearity analyses of *Paulinella* organelle genomes. (a) Collinearity analysis of chromatophore genome structures across phototrophic *Paulinella*, showing structural rearrangements. (b) Collinearity analysis of mitochondrial genome structures across phototrophic *Paulinella*, showing structural rearrangements. For both: Links are shaded by sequence identity (darker indicates higher identity). Brackish water *P. murrayi* sp. nov. (Cell 5) and *P. marae* (Cell 9) described in this study are highlighted in red.

979,356 bp in *P. longichromatophora*, and 1,021,616 bp in *P. chromatophora* CCAC0185 (Lhee et al., 2019). Gene content was broadly conserved: Cell 5 contained 866 CDSs, six rRNAs, and 42 tRNAs, whereas Cell 9 contained 861 CDSs, six rRNAs, and 42 tRNAs (Tables S5, S6). Gene content was further examined from the perspective of endosymbiotic gene transfer (EGT) candidate genes. Notably, specific differences were observed in the presence and absence of two EGT candidate genes. The PCC 0564 homolog, previously described as a putative *psaI* homolog (Lhee et al., 2019), was detected in the chromatophore genomes of *P. chromatophora* and *P. longichromatophora* but was not detected in the other phototrophic *Paulinella* chromatophore genomes, including those of Cells 5 and 9. In contrast, *psaK* was not detected in the chromatophore genome of *P. chromatophora* but was detected in the chromatophore genomes of the other phototrophic *Paulinella* species examined (Table S7). For the mitochondrial genome, Cell 5 yielded a partial assembly of 37,216 bp (i.e., three linear scaffolds), whereas Cell 9 yielded a circular 43,976 bp mitochondrial genome. Although the mitochondrial genome assembly of Cell 5 is incomplete, the mitochondrial gene content of Cell 9 is similar to that of other photosynthetic *Paulinella* mitochondrial genomes (Cell 5: 16 CDSs, two rRNAs, and 13 tRNAs; Cell 9: 19 CDSs, two rRNAs, and 22 tRNAs; Table S7). Despite these conserved overall genome features, structural variation was detected between Cells 5 and 9 in the chromatophore genomes, where a single large inversion was observed (Figure 3a). Moreover, many rearrangements were observed in the mitochondrial genomes (Figure 3b).

Inclusion of two novel species reveals phylogenetic conflict

Phylogenetic analyses revealed discordance between the nuclear and organelle phylogenies of photosynthetic *Paulinella*. In the 18S rDNA gene tree, *P. murrayi* (Cell 5) and *P. marae* (Cell 9) were paraphyletic, and *P. micropora* diverged after *P. chromatophora* (Figure 4). In contrast, in the phylogeny reconstructed from concatenated chromatophore genes, *P. murrayi* and *P. marae* formed a monophyletic clade, and *P. micropora* was the most recently diverging lineage within photosynthetic *Paulinella* (Figure 5), with strong bootstrap support (100%) for all relevant internal branches. In addition, a mitochondrial phylogeny

reconstructed from a concatenated dataset of 19 mitochondrial CDSs was congruent with the chromatophore phylogeny vis-à-vis the photosynthetic *Paulinella* clade, although bootstrap support for the ancestral position of *P. chromatophora* was relatively low (62%; Figure 6).

To assess phylogenetic conflict within photosynthetic *Paulinella*, a phylogeny was inferred using the nuclear 28S rDNA gene. In this case, the resulting topology not only differed from the 18S rDNA gene tree but also conflicted with the organelle phylogenies, with low bootstrap support values (Figure S11). These results suggest that a single marker gene, such as the 18S rDNA gene, may not be sufficient to resolve *Paulinella* phylogenetic relationships, especially as more species are discovered. Next, bipartitions from individual gene trees were mapped onto each clade of the concatenated chromatophore and mitochondrial phylogenies, and for each clade, the numbers of genes concordant with or conflicting with the corresponding split were summarized. In the chromatophore phylogeny, the monophyly of photosynthetic *Paulinella* was supported by most genes (concordant = 340; conflicting = 13; Figure S12). However, within photosynthetic *Paulinella*, conflict was relatively elevated at specific internal branches. The clade defined as *P. micropora* + *P. longichromatophora* + *P. murrayi* + *P. marae* was supported by 290 genes, whereas 61 genes supported alternative splits conflicting with this clade. Similarly, for the *P. micropora* + *P. longichromatophora* clade, 289 genes were concordant, and 61 were conflicting, confirming substantial gene-tree discordance within photosynthetic *Paulinella* (Table S8). For the concatenated mitochondrial phylogeny, all genes supported the monophyly of *Paulinella*, including heterotrophic taxa; however, extensive conflict was detected within *Paulinella* in internal relationships (Figure S13). These organelle gene trees were grouped into four major topologies within photosynthetic *Paulinella*, according to the number of genes supporting each topology (Figure 7). All four major topologies consistently placed *P. chromatophora* as the earliest-diverging lineage, and none of them matched the topology recovered from the 18S rDNA gene phylogeny. Chromatophore gene trees most frequently supported Type 1, the topology inferred using the concatenated chromatophore phylogeny ($n = 234$). Notably, the *rbcL* gene analysis, which is widely used as a conserved phylogenetic marker, supported Type 4 (Table S8). For the mitochondrial gene trees, only two genes supported Type

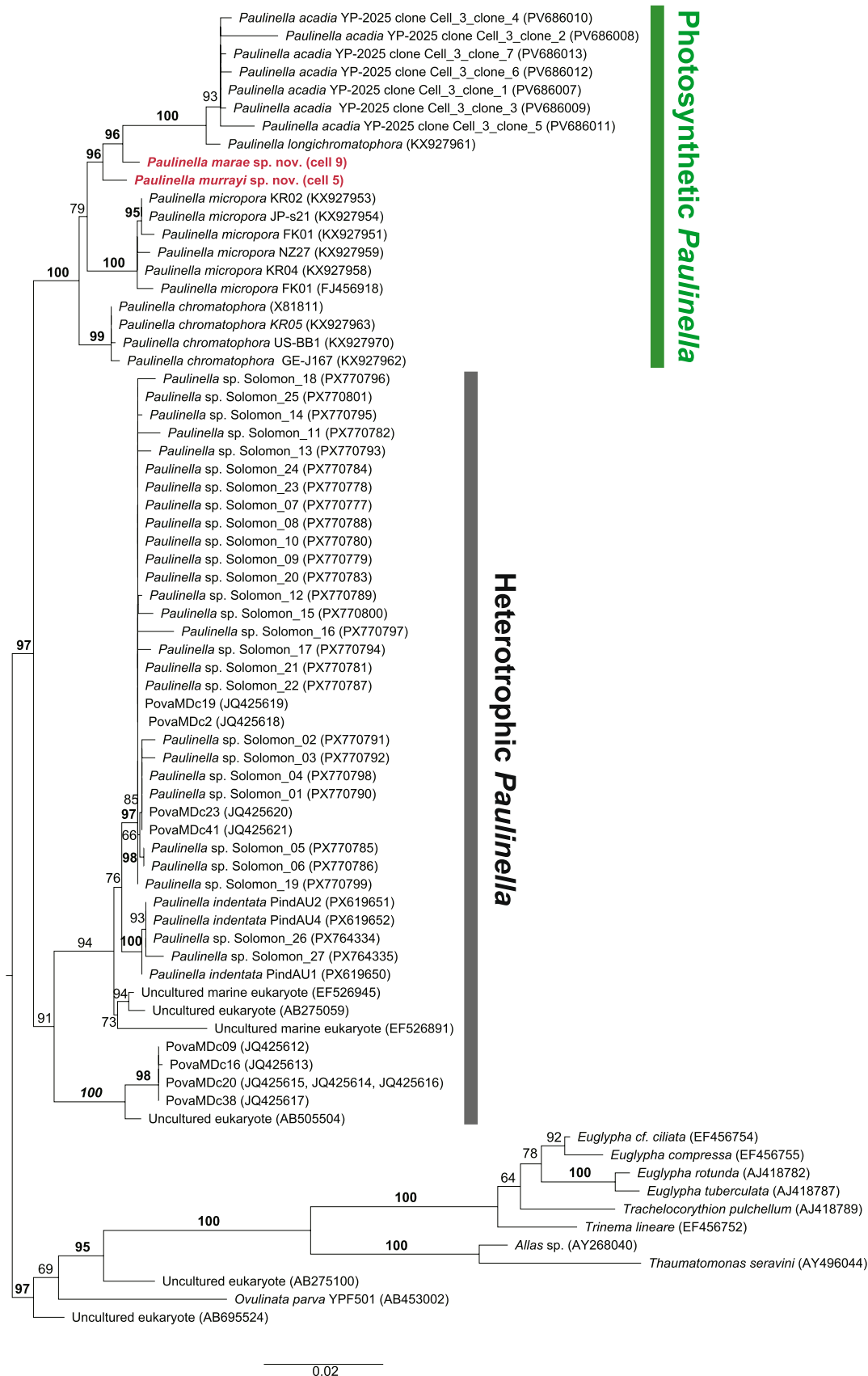


FIGURE 4 18S rDNA-based phylogenetic relationships of *Paulinella*. The maximum-likelihood tree was inferred from 18S rRNA gene sequences. *Paulinella murrayi* sp. nov. (Cell 5) and *P. marae* (Cell 9) are highlighted in red. Node values indicate ultrafast bootstrap (UFBoot) support values, and UFBoot values $\geq 95\%$ are shown in bold.

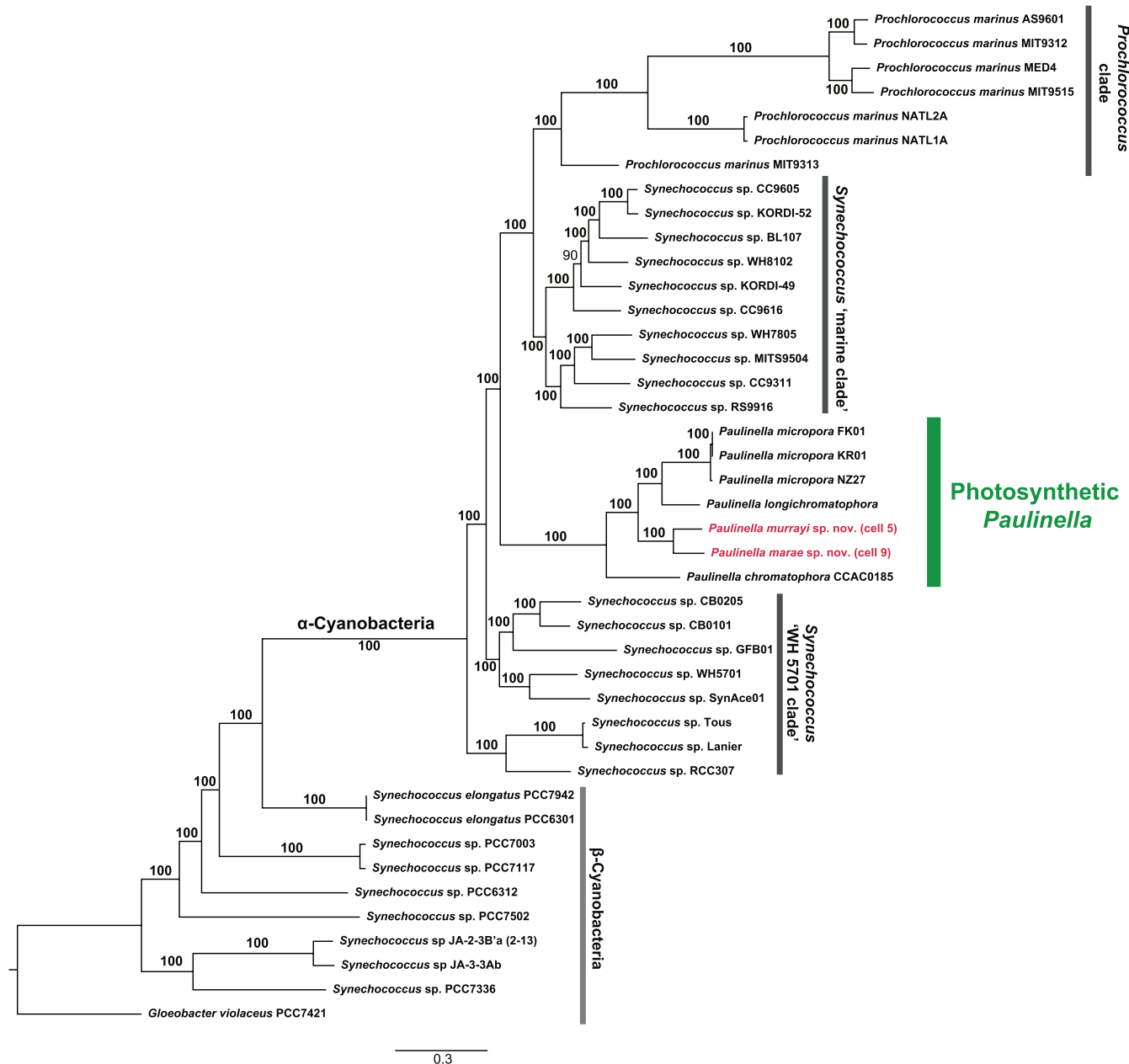


FIGURE 5 Chromatophore concatenated phylogeny of *Paulinella*, including representative cyanobacterial genomes, based on 406 genes. The maximum-likelihood tree was inferred from a concatenated amino acid alignment of 406 chromatophore-encoded proteins. *Paulinella murrayi* sp. nov. (Cell 5) and *P. marae* (Cell 9) are highlighted in red. Node values indicate ultrafast bootstrap (UFBoot) support values, and UFBoot values $\geq 95\%$ are shown in bold.

1, which may reflect generally low bootstrap support and frequent placements of heterotrophic *Paulinella* lineages within the photosynthetic *Paulinella* clade. Overall, gene-tree conflict was observed within photosynthetic *Paulinella*. Nevertheless, Type 1 was recovered by the concatenated chromatophore phylogeny (406 genes) with high bootstrap support, and the concatenated mitochondrial phylogeny also recovered Type 1. In addition, Type 1 was the most frequently observed topology among individual chromatophore gene trees.

The *rbcl* gene exhibits substitution saturation

Phylogenetic discordance analysis revealed that the RbcL amino acid tree recovered topology Type 4, which differed from the topology Type 2 found in the *rbcl* gene nucleotide tree used for species identification in photosynthetic *Paulinella* (Table S8; Figure S14). To evaluate whether this incongruence between nucleotide and amino acid-based phylogenies was associated with substitution saturation in the *rbcl* gene,

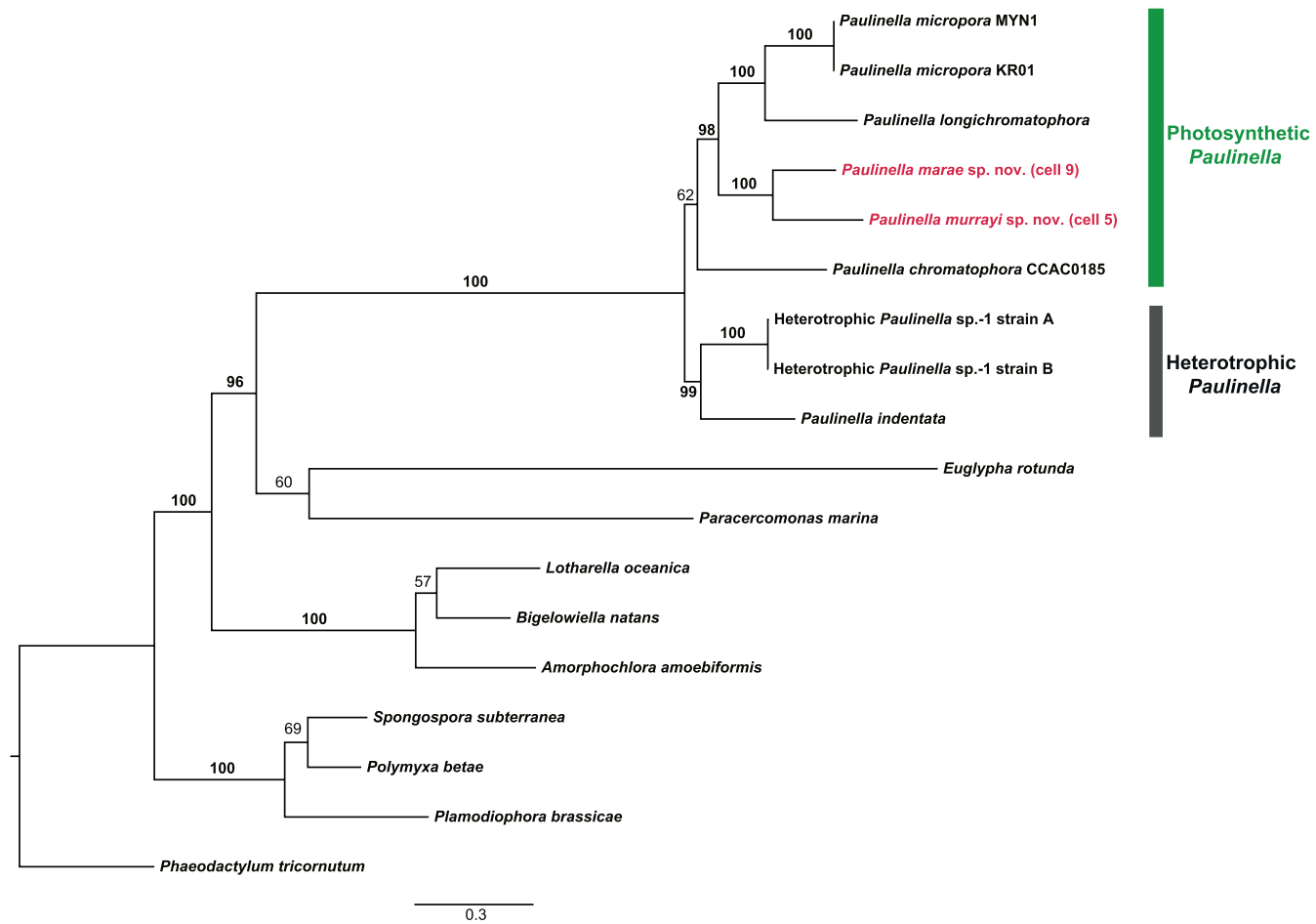


FIGURE 6 Mitochondrial concatenated phylogeny of *Paulinella*, including Rhizaria taxa and stramenopiles, based on 19 genes. The maximum-likelihood tree was inferred from a concatenated amino acid alignment of 19 mitochondrion-encoded proteins. *Paulinella murrayi* sp. nov. (Cell 5) and *P. marae* (Cell 9) are highlighted in red. Node values indicate ultrafast bootstrap (UFBoot) support values, and UFBoot values $\geq 95\%$ are shown in bold.

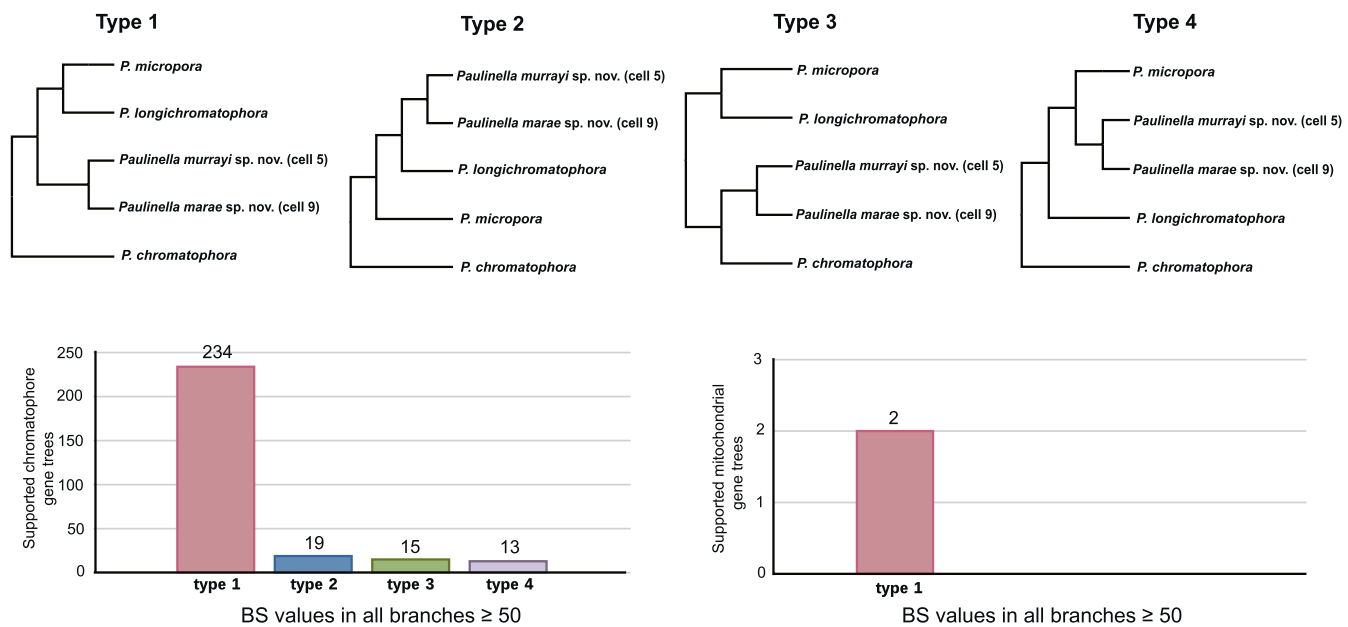


FIGURE 7 Major gene tree topologies among phototrophic *Paulinella* taxa. Four predominant topologies (Types 1–4) recovered from chromatophore and mitochondrial gene trees are shown, with bars indicating the number of supported gene trees for each topology (all branches with bootstrap support ≥ 50).

phylogenies were reconstructed for each codon position and saturation analyses were performed (Figure S14). Because substitution patterns are known to differ among codon positions, with second codon positions being the most conserved followed by first positions, and third codon positions evolving most rapidly (Xia et al., 2003), phylogenies were generated from datasets including all codon positions (NT123), the first and second codon positions only (NT12), and the second codon positions only (NT2). The NT123- and NT12-based phylogenies resulted in topologies that differed from the amino acid phylogeny, whereas the NT2-based phylogeny was congruent with the amino acid topology (Figure S14a–d).

Substitution saturation was assessed by comparing observed distances calculated directly from the alignment with inferred distances estimated from the maximum-likelihood tree (Philippe & Forterre, 1999). In the *rbcL* gene, inferred substitutions exceeded observed substitutions more strongly in NT123 than in the amino acid dataset, with the greatest excess observed at the third codon position (NT3; Figure S14e,f). These results indicate that substitution saturation occurs in the *rbcL* gene, particularly at third codon positions, which fits the expected model.

DISCUSSION

Possible causes of phylogenetic incongruence in photosynthetic *Paulinella*

Understanding phylogenetic relationships within photosynthetic *Paulinella* may provide key insight into the ecology of the ancestor that experienced the primary endosymbiosis event. However, the inclusion of *P. murrayi* (Cell 5) and *P. marae* (Cell 9) in the phylogenetic analyses revealed phylogenetic incongruence among taxa. Several factors may explain this result. First, the 18S rDNA, 28S rDNA, and *rbcL* genes are insufficient, as single genes, to resolve taxon interrelationships. (For examples in other lineages, see Theriot et al., 2009; De Luca et al., 2021.) Therefore, the incongruence observed here between the single gene-based phylogenies and the organelle phylogenies inferred from concatenated sequences may reflect the more limited phylogenetic signal in the former.

Second, substitution saturation may have affected some genes, which can mislead phylogenetic inference (Xia et al., 2003). In this study, the *rbcL* gene analysis results suggest that substitution saturation affected some codon positions, particularly the third codon position, and may thereby have contributed to phylogenetic incongruence. This phenomenon may have impacted other chromatophore-encoded genes in *Paulinella*, resulting in phylogenetic conflict.

Third, current taxon-sampling may be insufficient for robust reconstruction of phylogenetic relationships within *Paulinella*. In general, limited taxon sampling can reduce the accuracy of phylogenetic inference, and the absence of intermediate lineages may increase the likelihood of long-branch attraction or unstable branching patterns (Nabhan & Sarkar, 2012; Zwickl & Hillis, 2002). Resolving this phylogenetic conflict will require denser taxon sampling together with broader genomic and geographic coverage across under-sampled *Paulinella* lineages. In particular, whole-genome data and multigene analyses incorporating both nuclear and organelle loci may be necessary to reconstruct the *Paulinella* phylogeny with greater confidence. As phylogenetic resolution improves, current species designations and clade boundaries may need to be reevaluated before ecological transitions and ancestral character states are inferred with confidence.

Growth in knowledge about the low-abundant but widespread photosynthetic *Paulinella*

The rarity of known photosynthetic *Paulinella* populations, coupled with their slow growth rate and bleaching (loss of chlorophyll) under even moderate light levels (Zhang et al., 2017), has led to the perception that this organism may not be well adapted to its environment and is a peculiarity rather than an evolutionary success story, that is, from the human perspective. However, as we develop a more sophisticated understanding of the ecology of members of this lineage and their habitat range and distributions, we reach more informed conclusions about the evolutionary strategies that have allowed *Paulinella* to persist for over 100 mya, during which slow growth and low abundance may have been beneficial. *Paulinella chromatophora* was first described by Lauterborn (1895) from sediments of the Rhine River (freshwater habitat, Germany) and was isolated from a nearby location in 1990 (<https://www.uni-due.de/biology/ccac/details.php?strain=CCAC%200185>). *Paulinella micropora* also inhabits freshwater and has been isolated from sites in South Korea, Europe, Japan, and New Zealand (see table S5 in Lhee et al., 2019). *Paulinella longichromatophora* was isolated from a beach in South Korea and is the only described marine species (Kim & Park, 2016). *Paulinella acadia* (for which only ribosomal sequence data exist; Pardasani et al., 2025), and *P. murrayi* and *P. marae* were observed in brackish water habitats (with different salinity levels) on opposite coasts of North America. There are also nine heterotrophic sister species described (Han et al., 2026; Johnson et al., 1988; Nicholls, 2009), but none are yet available in culture collections; only one species has an available single-cell genome (*P. ovalis*;

Bhattacharya et al., 2012), and the taxonomic assignments are still being clarified and revised (e.g., Han et al., 2026; Yoon lab, personal communication). These scientific efforts have been complemented by a growing number of observations by naturalists (Figures S1, S2) and DNA barcoding efforts (Figure S2). In fact, because of hobby scientists, there have been more documented sightings of *Paulinella* spp. in new locations in the last 5 years than in the previous 125 years (Figures S1, S2), signifying a cosmopolitan distribution of these photosynthetic species.

Finding two distinct species in one environment presents the unique opportunity to investigate the ecological role, distribution, and population genetics of photosynthetic *Paulinella*. *Paulinella murrayi* and *P. marae* appear to be sister species (Figures 5 and 6); both are likely present at various points along the east coast of the United States, and their populations overlap in the Nag's Head, North Carolina region. These results highlight the need for expanded sampling efforts that should be informed by iNaturalist observations, Global Biodiversity Information Facility data, and historical accounts (e.g., Kepner, 1905; Lackey, 1936). There is value in slow and steady naturalist work to discriminate between populations, and this situation offers the opportunity for hobbyists and amateur microscopists to contribute to research by collaborating with scientists. Without this kind of observational legwork on natural populations (i.e., if we had sampled the first cells detected in a natural setting, established a clonal culture, and moved on with genomics), we would not have noticed that there were two morphologies and ultimately two species at the North Carolina location that displayed different behaviors, which allowed for the formulation of hypotheses about their ecology and evolutionary history. For *P. marae*, we hypothesize an epiphytic lifestyle, that is, the filopodia used for movement and once used to feed in the ancestral heterotrophic state (Figure S9) have been co-opted for adhesion, perhaps keeping this smoother-scaled species sessile and attached to plants or diatom sheaths, at the optimal light intensity (Figure S8). In contrast, in *P. murrayi*, these smaller, outwardly more robust cells are typically observed free-floating. Perhaps rougher texture of the scales and more compact structure may be an adaptation to light conditions or challenges with free-floating at the water's surface (Figures 1, S5, S6). These ideas are still speculative but are now possible to investigate going forward.

Endosymbiotic ratchet as a driver of *Paulinella* divergence

We previously proposed the “endosymbiotic ratchet hypothesis” (ERH) as a provisional framework for understanding the outcomes of post-plastid primary endosymbiosis and its impacts on cell evolution (Bhattacharya et al., 2023). The ERH was built on a single model,

Paulinella, given the paucity of other datasets that address primary plastid origin, and it is therefore a working hypothesis in need of testing. Despite its narrow origin story, the ERH is mechanistically grounded in established population genetics. Specifically, the effective population size–drift–genome–size relationship is well supported in independent literature (Charlesworth, 2009; Lynch, 2020) and the impact of Muller's ratchet on non-recombining endosymbiont genomes is also well understood (Muller 1964). Essentially, the ERH states that primary endosymbiosis acts as an irreversible, unidirectional driver of rapid lineage divergence and evolutionary complexity (Bhattacharya et al., 2023). A newly formed symbiotic relationship will putatively limit effective population size and increase reproductive isolation and generation times, accelerating genetic differentiation by increasing the effects of genetic drift and relaxed selection (Bhattacharya et al., 2023; Han et al., 2026). Unfortunately, no direct measurements of population size exist for wild *Paulinella*, and the size of ancestral populations post-endosymbiosis must (necessarily) be inferred. Nonetheless, these processes could explain both why there is a 10-fold genome size difference between *P. micropora* and *P. chromatophora* and the presence of morphologically distinct, small populations worldwide (Bhattacharya et al., 2023). The effects of such a significant evolutionary transition are expected to also leave “footprints” on organellar genomes. For example, relaxed selection on mitochondrial genomes can result from a bottleneck that dramatically reduces effective population size (e.g., after a primary endosymbiotic event), even if selective constraints on individual mitochondrial genes are strong (Han et al., 2026). In conjunction with the evolution of a new organelle (e.g., the chromatophore) that slowly integrates with the host genome, generation times of the organisms will increase, keeping populations smaller in size, which we see each time a new phototrophic *Paulinella* species is reported in nature and is invariably in very low abundance. This phenomenon impacts mitochondrial genome structure through increased gene order changes via translocations that are mildly deleterious but fixed due to lowered levels of selection in small populations (Han et al., 2026). Figure 3b shows evidence for this idea, and we can also observe that within a species (at the strain level, e.g., between *P. micropora* MYN1 and KR01), mitochondrial gene order is preserved more than between species. Comparatively, the degree of rearrangement between *P. marae* and *P. murrayi* appears to be indicative of species-level divergence. Moreover, the chromatophore genomes are under greater selection for maintaining gene order, suggesting that this organelle is still a “work in progress” and may be in a period of punctuated equilibrium between major waves of genome reduction and EGT (Calatrava et al., 2024; Gabr et al., 2022; Nowack et al., 2011, 2016). This is also evidence (Figure 3a) that there is a lower degree

of rearrangement across all species and only one major structural rearrangement between *P. marae* and *P. murrayi*. This result suggests that although these species diverged some time ago, they are more closely related to each other than to other known species. This observation supports the idea that there may be many uncharacterized species across the planet, and addition of their genomic data to these analyses will clarify phylogenetic relationships and help further develop evolutionary hypotheses.

CONCLUSIONS AND FUTURE DIRECTIONS

By identifying and describing *Paulinella marae* and *P. murrayi*, we have expanded the known diversity of this clade from four to six species and described an easy-to-access field site with co-occurring species reliably observed during each sampling endeavor. These two species may have diverged while in sympatry due to micro-habitat adaptations that drove speciation. Alternatively, *P. marae* and *P. murrayi* may have arisen in different habitats and were later united due to range shifts, thereby explaining their co-occurrence. We currently do not yet have the data to distinguish between these two competing scenarios. The North Carolina site is nevertheless ideal for future work to elucidate photosynthetic *Paulinella* spp. population genetics, species interactions, and other ecological features that will advance understanding of the implications of lifestyle transitions and outcomes of primary endosymbiosis. We hope that by presenting this research alongside the work of hobby scientists who have identified these enigmatic protists in their local ponds and beaches, acceleration of such efforts will result in the discovery of additional species, more cultures established for experimentation, and more data generated for comparative genomics and other approaches.

AUTHOR CONTRIBUTIONS

Julia Van Etten: Conceptualization (lead); data curation (equal); formal analysis (equal); funding acquisition (equal); investigation (lead); methodology (equal); project administration (lead); validation (equal); visualization (lead); writing – original draft (lead); writing – review and editing (lead). **Sungpil Han:** Data curation (lead); formal analysis (lead); investigation (equal); methodology (lead); validation (lead); visualization (lead); writing – original draft (equal); writing – review and editing (supporting). **John A. Burns:** Conceptualization (supporting); data curation (supporting); formal analysis (supporting); investigation (supporting); methodology (supporting); resources (supporting); software (supporting); supervision (supporting); validation (supporting); visualization (supporting). **Duckhyun Lhee:** Data curation (equal); formal analysis (equal); investigation

(supporting); methodology (equal); validation (equal); visualization (supporting); writing – review and editing (supporting). **Andrew C. Willoughby:** Investigation (supporting); validation (supporting); visualization (supporting); writing – review and editing (supporting). **Timothy G. Stephens:** Investigation (supporting); methodology (supporting); writing – review and editing (supporting). **Erin Chille:** Investigation (supporting); writing – review and editing (supporting). **Robin S. Sleith:** Data curation (supporting); formal analysis (supporting); investigation (supporting); validation (supporting). **Debashish Bhattacharya:** Conceptualization (equal); funding acquisition (equal); investigation (equal); methodology (equal); project administration (equal); resources (equal); supervision (equal); validation (equal); writing – review and editing (equal). **Hwan Su Yoon:** Conceptualization (equal); data curation (equal); funding acquisition (equal); investigation (lead); project administration (equal); resources (lead); supervision (lead); validation (equal); writing – review and editing (equal).

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DATA AVAILABILITY STATEMENT

All genomic data including chromatophore assemblies, mitochondrial assemblies, and 18S, 28S, and rbcL sequences can be found in NCBI under BioProject [PRJNA1453438](https://www.ncbi.nlm.nih.gov/bioproject/PRJNA1453438): Biosamples SAMN57262795 for *Paulinella marae*—cell 9 (TaxID: 3701295) with raw reads as [SRR38183186](https://www.ncbi.nlm.nih.gov/sra/SRR38183186), and SAMN57262796 for *Paulinella murrayi*—cell 5 (TaxID: 3701296) with raw reads as [SRR38183187](https://www.ncbi.nlm.nih.gov/sra/SRR38183187). Marker gene sequences—*P. marae*: 18S (PZ578613), 28S (PZ582844), and rbcL (PZ584713); *P. murrayi*: 18S (PZ578619), 28S (PZ582845), and rbcL (PZ584714). *P. marae* ZooBank accession: urn:lsid:zoobank.org:act:20B9C7A3-B9FB-4B0E-9E1E-299EC796D480; *P. murrayi* ZooBank

accession: urn:lsid:zoobank.org:act:904668FC-5B5F-4DA5-B987-25BDBB92312B. Type specimens are deposited in the Chrysler Herbarium under accession no. CHRB0145024 for *P. marae* and CHRB0145025 for *P. murrayi*.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Figure S1. Global distribution of *Paulinella* spp. based on observational data from iNaturalist.

Figure S2. Global distribution of *Paulinella* spp. based on observational and DNA barcoding data from the Global Biodiversity Information Facility (GBIF).

Figure S3. Location information for collection of *P. marae* sp. nov. and *P. murrayi* sp. nov.

Figure S4. Micrographs from field samples showing two morphologies.

Figure S6. Inverted SEM image of *P. murrayi* sp. nov. showing that pores appear as thinner (less dense) parts of the scales and become less pronounced or visible with this type of imaging in the posterior scales, which have the raised punctae.

Figure S7. Handedness in *P. murrayi* sp. nov. vs. *P. marae* sp. nov. Scales in *P. murrayi* sp. nov. are

right-handed, overlapping in a clockwise direction (A). Scales in *P. marae* sp. nov. are left-handed, overlapping in the anti-clockwise direction (B).

Figure S8. Brightfield images from field samples of *P. marae* sp. nov. cells adhering to substrate.

Figure S9. Filopodia extensions to >4 times cell length in a *P. murrayi* cell (top, A) and *P. marae* cell (bottom, B). Phase contrast.

Figure S10. Link to *Paulinella* moving video that was unable to be uploaded within this document: <https://youtu.be/26cuS-jVom8>. Video of cell 2025_12 (*P. murrayi*) moving and extending its filopodia.

Figure S11. 28S rDNA-based phylogenetic relationships of *Paulinella*. The maximum-likelihood tree was inferred from 28S rRNA gene sequences. *P. murrayi* sp. nov. (cell 5) and *P. marae* (cell 9) are highlighted in red.

Figure S12. Summary of gene-tree discordance inferred by PhyParts for phototrophic *Paulinella*, using the chromatophore concatenated phylogeny as the reference tree.

Figure S13. Summary of gene-tree discordance inferred by PhyParts for phototrophic *Paulinella*, using the mitochondrial concatenated phylogeny as the reference tree.

Figure S14. Comparison of rbcL phylogenies and substitution saturation among photosynthetic *Paulinella* taxa.

Table S1. A record of the cells from this location that could be measured and their putative diagnosis based on morphological characters.

Table S2. Fates of cells isolated for WGA.

Table S3. Percent identity between marker genes indicates two novel species.

Table S4. Assembly statistics for organelle genomes.

Table S5. Chromatophore genome content.

Table S6. Mitochondrial genome content across phototrophic and heterotrophic *Paulinella* species.

Table S7. Comparison of chromatophore protein-coding gene content among phototrophic *Paulinella* species.

Table S8. ILS_table.

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